CipherGuard: AI-Powered Password Intelligence Platform

CipherGuard is a world-class, production-grade AI-Powered Password Intelligence Platform built with a focus on security, privacy, and user empowerment. Every analysis is performed locally in your browser with **zero logging** or data transmission.

🌟 What's New in Enhanced Edition (v2.0)

✨ New Features

- **Strength History Chart** - Visual entropy trends showing password strength evolution over time
- **Breach Database Simulator** - Check your password against 1000+ known compromised passwords
- **Smart Recommendations Engine** - AI-like suggestions for password improvement with actionable insights

🔐 Core Features

- **Real-time Entropy Calculation** - Shannon's formula for accurate entropy scoring ($E = \log_2(R^L)$)
- **Advanced Pattern Detection** - Keyboard walks, sequences, dictionary matches, leet-speak decoding
- **Cryptographically Secure Generation** - Multiple modes (Maximum Security, Memorable Passphrase)
- **Dynamic Security Grading** - Letter grades A+ to F with detailed assessments
- **Crack Time Estimator** - Comparative table for Offline Fast (100B/s), Online Throttled (1k/s), GPU Cluster (1T/s)
- **100% Offline-Capable** - No external APIs, frameworks, or dependencies

🎯 Key Highlights

Security Architecture

- ✅ **Zero External Dependencies** - Pure HTML5, CSS3, Vanilla JavaScript
- ✅ **Volatile Memory Only** - All data cleared on page close
- ✅ **Cryptographic Random** - Uses `crypto.getRandomValues()` for secure generation
- ✅ **Sanitized Inputs** - All user inputs validated before processing
- ✅ **No Network Calls** - 100% offline operation
- ✅ **Screen Blur Security** - Automatic blur when window loses focus
- ✅ **Auto-Clear Timer** - 60-second auto-clear of sensitive data

Visual Identity: "Stranger OS" Aesthetic

- **Color Palette:** #0a0a0a (BG), #e8ff00 (Electric Yellow), #00ff88 (Success), #ff4444 (Danger)
- **Typography:** JetBrains Mono, 12px border-radius, 1px borders
- **Effects:** Matrix-rain canvas, scanline overlay, subtle noise texture
- **Micro-interactions:** Glowing inputs, smooth transitions, responsive feedback

📊 Features in Detail

1. Smart Analyzer

- Real-time entropy calculation as you type
- Pattern detection (uppercase, lowercase, numbers, symbols)
- Keyboard walk detection (qwerty, asdfgh, etc.)
- Dictionary word matching
- Sequence detection (abc, 123, etc.)
- 5-bar strength meter with color coding
- Comparative crack-time table

2. Pro Generator

- **Maximum Security Mode:** Cryptographically random passwords
- **Memorable Mode:** Passphrase-based passwords (e.g., "quick-fox-jumps-742")
- Configurable length (8-128 characters)
- Character set options (uppercase, lowercase, numbers, symbols)
- One-click copy to clipboard
- Real-time entropy display

3. Breach Database Check (NEW)

- Check password against 1000+ known compromised passwords
- Local analysis - password never leaves your browser
- Instant results with security status
- Recommendations if password is found in breach database

4. Smart Recommendations (NEW)

- AI-like suggestions for password improvement
- Specific actionable recommendations
- Real-time updates as you type
- Covers length, character diversity, patterns, dictionary words

5. Strength History Chart (NEW)

- Visual representation of last 10 password entropy calculations
- Bar chart showing entropy trends
- Interactive bars with hover information

- Helps track password improvement over time

6. Security Report

- Dynamic letter grade (A+ to F)
- Comprehensive security assessment
- Entropy score, length, character diversity analysis
- Warning flags for keyboard walks and dictionary matches
- Breach database status
- One-click export to clipboard

🚀 Getting Started

Installation

No installation required! Simply open the HTML file in any modern web browser:

```bash
# Download the file
curl -O https://raw.githubusercontent.com/mr-usmanrouf/CipherGuard/main/CipherG

# Open in browser
open CipherGuard_Enhanced.html
# or
firefox CipherGuard_Enhanced.html
# or
chrome CipherGuard_Enhanced.html
```

Usage

Analyze a Password

1. Click the **Analyzer** tab
2. Type your password in the input field
3. Watch real-time entropy calculation, pattern detection, and recommendations
4. View strength history chart and crack time estimates

Generate a Secure Password

1. Click the **Generator** tab
2. Choose generation mode (Maximum Security or Memorable)
3. Configure character set and length
4. Click "Generate Password"
5. Click "Copy" to copy to clipboard

Check Breach Database

1. Click the **Breach Check** tab
2. Enter a password to check
3. Click "Check Against Database"
4. See if password appears in known breaches

Generate Security Report

1. Click the **Report** tab
2. Enter a password
3. Click "Generate Report"
4. View detailed security assessment with letter grade
5. Click "Export Summary" to copy report to clipboard

📈 Technical Specifications

Entropy Calculation

Plain Text

```
E = log₂(charset_size ^ password_length)

Where:
- charset_size = number of possible characters
- password_length = length of password
```

Crack Time Formula

Plain Text

```
Time = (2^entropy / 2) / attack_speed_per_second
```

Security Grading Algorithm

- **40%** - Entropy score
- **20%** - Password length
- **20%** - Character diversity
- **20%** - Penalties (keyboard walks, sequences, dictionary words)

Grade Scale

Grade	Score	Assessment
A+	90-100	Excellent - Highly Secure
A	85-89	Excellent - Very Secure
B+	75-84	Good - Reasonably Secure
C	50-74	Fair - Needs Improvement
D	40-49	Weak - Vulnerable
F	0-39	Very Weak - Highly Vulnerable

🔒 Privacy & Security

CipherGuard is designed with privacy as the top priority:

- ✅ **Zero-Logging Architecture** - No passwords or data stored
- ✅ **Local Analysis Only** - All processing happens in your browser
- ✅ **No External Requests** - No API calls or data transmission
- ✅ **Volatile Memory** - Data cleared on page close
- ✅ **Open Source** - Audit the code yourself
- ✅ **No Tracking** - No analytics or telemetry

🛠️ Technology Stack

- **Frontend:** HTML5, CSS3 (Flexbox/Grid), Vanilla JavaScript (ES6+)

- **Fonts:** JetBrains Mono (Google Fonts)
- **Cryptography:** Native `crypto.getRandomValues()` API
- **Architecture:** Single-file deployment, zero dependencies
- **Responsive:** Mobile-first design, works on all devices

📱 Browser Compatibility

- ✅ Chrome/Chromium 90+
- ✅ Firefox 88+
- ✅ Safari 14+
- ✅ Edge 90+
- ✅ Mobile browsers (iOS Safari, Chrome Mobile)

📚 File Structure

Plain Text

```
CipherGuard/
├── CipherGuard_Enhanced.html    # Main application (single file)
├── README.md                     # This file
├── LICENSE                       # MIT License
├── CHANGELOG.md                  # Version history
└── CONTRIBUTING.md               # Contribution guidelines
```

🎨 Customization

The application uses CSS variables for easy customization:

CSS

```
:root {
    --bg-primary: #0a0a0a;
    --accent-yellow: #e8ff00;
    --accent-green: #00ff88;
    --accent-red: #ff4444;
    --accent-cyan: #00ffff;
}
```

Edit these values in the `<style>` section to customize colors.

🐛 Known Limitations

- Breach database contains 1000+ most common passwords (not exhaustive)
- For production use, integrate with real breach databases (Have I Been Pwned API)
- Password recommendations are rule-based, not AI-powered
- Entropy calculation assumes random password generation

🚀 Future Enhancements

- [ ] Integration with Have I Been Pwned API (with user consent)
- [ ] Machine learning-based password strength prediction
- [ ] Multi-language support
- [ ] Dark/Light theme toggle
- [ ] Password history tracking (local storage, optional)
- [ ] Keyboard shortcuts
- [ ] Export reports as PDF
- [ ] Mobile app version

📄 License

MIT License - See LICENSE file for details

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🤝 Contributing

Contributions are welcome! Please feel free to submit a Pull Request.

📞 Support

For issues, questions, or suggestions:

1. Check existing issues on GitHub
2. Create a new issue with detailed description
3. Contact via Instagram: @mr.usmanrouf

🙏 Acknowledgments

- JetBrains for the excellent Mono font
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- The cybersecurity community for inspiration
- All contributors and users

⭐ Show Your Support

If you find CipherGuard useful, please:

- ⭐ Star this repository
- 🔗 Share with friends and colleagues
- 💬 Leave feedback and suggestions
- 🐛 Report bugs and issues

Built with ❤️ by Muhammad Usman Rouf

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"Security is not a feature, it's a responsibility."